

SUPPLY MANAGEMENT SYSTEM — Functional Specification Document

FSD Addendum 32 — Sales Pre-Order Pipeline (Inquiry → Quotation → Sale Order)

Property	Value
Document ID	SMS-FSD-ADD-032
Version	1.0
Date	2026-10-03
Status	Draft
Author	System Architect
Classification	Internal — Development
Supersedes	None — new entities extending demand schema
Architecture	.NET 8 / EF Core 8 / SQL Server 2022 / React 18
Change Items	5 new features & extensions
Impact	7 migrations, 6 new entities, 2 modified entities, 1 new lookup, 5 new UI panels, 3 state machines
Deferred to Addendum 33	Fulfillment orchestration, delivery notes, shipment tracking, invoice generation

Table of Contents

- 1. [Purpose & Scope](#)
- 2. [Pre-Order Pipeline Overview](#)
- 3. [C1 — Sale Inquiry](#)
- 4. [C2 — Sale Quotation](#)
- 5. [C3 — Sale Order Extensions \(Source Linking & Customer PO Capture\)](#)
- 6. [C4 — Sale Order Line Delivery Indicators & Inventory Reservation](#)
- 7. [C5 — Rejection Reason Lookup](#)
- 8. [Database Migrations](#)
- 9. [API Changes](#)

- 10. UI Wireframes & Specifications
 - 11. Business Rules
 - 12. Test Scenarios
 - 13. Development Phases
-

1. Purpose & Scope

This addendum introduces the Sales Pre-Order Pipeline — a three-document chain (Sale Inquiry → Sale Quotation → Sale Order) that covers the seller-side sales process from initial customer inquiry through to a confirmed order with inventory reservation. Both Sale Inquiry and Sale Quotation are OPTIONAL steps; a Sale Order may be created directly without either predecessor.

This addendum does NOT cover fulfillment, delivery notes, shipment tracking, or invoicing — those are deferred to Addendum 33.

1.1 Design Principles

- **Optional Chain, Maximum Traceability** — Inquiry and Quotation are optional steps, but when used the full chain is linked (Inquiry → Quotation → Sale Order) with forward and backward FK references for end-to-end traceability.
- **Line-Level Granularity** — Every evaluation, rejection, alternative, and customer response is tracked per line item, not per document. This enables fine-grained analytics on why products are rejected and which alternatives customers prefer.
- **Analytics-Ready Rejection Tracking** — Inquiry line rejections are captured with a formal RejectionReason lookup. This feeds future dashboards answering: "How many inquiries did we decline per reason, per product, per quarter?"
- **Separation of Concerns** — Sale Quotation (seller-initiated pricing proposal) is entirely separate from the existing RFQ implementation (buyer-initiated request for pricing). They are different document types with different schemas and workflows.
- **RBAC-Controlled Reservation** — Inventory reservation on Sale Order lines requires explicit permission. The reserve action is disabled in the UI unless the logged-in user holds the `sales_reserve_inventory` claim. Releasing reserved stock follows the same permission model.
- **Multi-Tenant by Default** — Every new entity carries `org_id` with EF Core `HasQueryFilter`. No exception.

1.2 Scope Summary

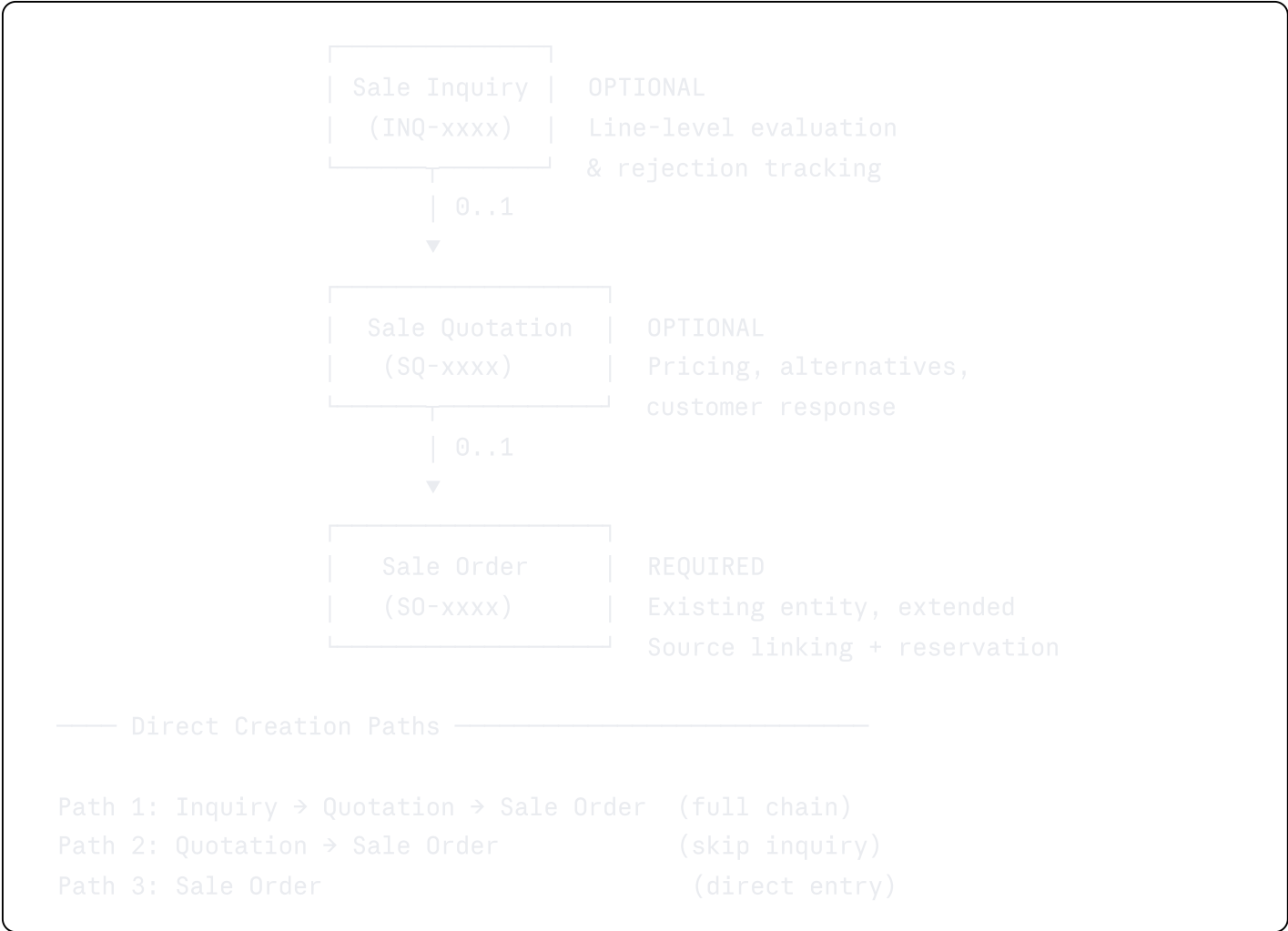
Change ID	Title	Type	Impact
C1	Sale Inquiry	New Entity + UI	2 new tables (SaleInquiries, SaleInquiryLines), 1 state machine, 1 new UI panel
C2	Sale Quotation	New Entity + UI	2 new tables (SaleQuotations, SaleQuotationLines), 1 state machine, 1 new UI panel
C3	Sale Order Extensions	Schema Extension	Add source linking + customer PO capture columns to existing SaleOrders
C4	SO Line Delivery Indicators & Reservation	Feature Extension	Line-level status indicators (green/yellow/red) + RBAC reservation column on existing SaleOrderLines
C5	Rejection Reason Lookup	New Lookup	1 new table (RejectionReasons) with seed data for analytics

1.3 Relationship to Existing Entities

Existing Entity	Schema	Addendum 32 Interaction
demand.SaleOrders	demand	EXTENDED — add source_quotation_id, source_inquiry_id, source_type, customer_po_reference, customer_po_date, customer_po_attachment_id
demand.SaleOrderLines	demand	EXTENDED — add reserved_qty column; delivery indicator is computed, not stored
demand.SaleOrderConfig	demand	UNCHANGED — reservation_ttl_hours already exists and applies
suppliers.BusinessPartners	suppliers	REFERENCED — SaleInquiry/Quotation link to customer via partner_id (partner_type includes CUSTOMER)
inventory.StockReservations	inventory	USED — IStockReservationService.Reserve/Release with source_type=SALES_ORDER
lookups.Products	lookups	REFERENCED — Inquiry/Quotation lines reference product variants

2. Pre-Order Pipeline Overview

2.1 Three-Document Chain



2.2 Document Numbering

Document	Pattern	Example	Auto-Generated
Sale Inquiry	INQ-{YYYY}-{NNNN}	INQ-2026-0001	Yes — per org, per year, sequential
Sale Quotation	SQ-{YYYY}-{NNNN}	SQ-2026-0001	Yes — per org, per year, sequential
Sale Order	SO-{NNNN}	SO-0001	Yes — existing pattern unchanged

3. C1 — Sale Inquiry

3.1 Overview

A Sale Inquiry captures an inbound customer request — received by email, phone, portal, or any other channel — and allows the sales team to evaluate each requested line item individually. The

inquiry is the only place where line-level evaluation (CAN_SUPPLY, CANNOT_SUPPLY, etc.) and rejection reasons are captured, providing analytics data for future product and capacity decisions.

 **NOTE:** Sale Inquiry is an OPTIONAL step. Organizations that do not need inquiry tracking can skip directly to Sale Quotation or Sale Order creation.

3.2 Schema — SaleInquiries

New Table: `demand.SaleInquiries`

Column	Type	Nullable	Default	Description
sale_inquiry_id	BIGINT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator — EF Core HasQueryFilter
inquiry_number	NVARCHAR(20)	NOT NULL	Auto	Pattern: INQ-{YYYYY}-{NNNN}, unique per org
customer_id	BIGINT	NOT NULL	FK → BusinessPartners	Customer who sent the inquiry
customer_reference	NVARCHAR(50)	NULL		Customer's own reference number (their RFQ number, email subject, etc.)
customer_reference_date	DATE	NULL		Date on the customer's reference document
status	NVARCHAR(20)	NOT NULL	'RECEIVED'	State machine: RECEIVED → UNDER_REVIEW → REVIEW_COMPLETE → QUOTED → DECLINED
received_date	DATE	NOT NULL	GETDATE()	Date inquiry was received
response_deadline	DATE	NULL		Customer-requested response date
assigned_to_user_id	BIGINT	NULL	FK → auth.Users	Sales rep assigned to evaluate
notes	NVARCHAR(2000)	NULL		Internal notes

Column	Type	Nullable	Default	Description
created_by	BIGINT	NOT NULL	FK → auth.Users	User who entered the inquiry
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
updated_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
trace_id	UNIQUEIDENTIFIER	NOT NULL	NEWID()	Correlation ID

Indexes:

Index	Columns	Type
UQ_SaleInquiries_OrgNumber	org_id, inquiry_number	Unique
IX_SaleInquiries_Customer	customer_id	Non-unique
IX_SaleInquiries_Status	org_id, status	Non-unique, filtered
IX_SaleInquiries_AssignedTo	assigned_to_user_id	Non-unique, filtered (WHERE NOT NULL)

3.3 Schema — SaleInquiryLines

New Table: `demand.SaleInquiryLines`

Column	Type	Nullable	Default	Description
inquiry_line_id	BIGINT IDENTITY	NOT NULL	PK	Primary key
sale_inquiry_id	BIGINT	NOT NULL	FK → SaleInquiries	Parent inquiry
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
line_number	INT	NOT NULL	Auto	Sequential within inquiry
product_id	BIGINT	NULL	FK → Products	Requested product (NULL when customer describes a non-catalog item)
variant_id	BIGINT	NULL	FK → ProductVariants	Specific variant if identified
product_description	NVARCHAR(500)	NOT NULL		Free text — customer's description (always captured, even when product_id is set)
requested_quantity	DECIMAL(18,4)	NOT NULL		Quantity the customer is requesting
requested_uom_id	INT	NULL	FK → UnitOfMeasures	Customer's requested unit of measure
requested_delivery_date	DATE	NULL		Customer's desired delivery date
line_status	NVARCHAR(20)	NOT NULL	'PENDING'	Evaluation status: PENDING, CAN_SUPPLY, PARTIAL, CANNOT_SUPPLY, UNDER_REVIEW
can_supply_quantity	DECIMAL(18,4)	NULL		Quantity the org can supply (when

Column	Type	Nullable	Default	Description
				PARTIAL)
estimated_delivery_date	DATE	NULL		Org's estimated delivery date for this line
rejection_reason_id	INT	NULL	FK → RejectionReasons	Why the line was rejected (when CANNOT_SUPPLY)
rejection_notes	NVARCHAR(500)	NULL		Additional rejection explanation
alternative_product_id	BIGINT	NULL	FK → Products	Suggested alternative product (when original rejected)
alternative_variant_id	BIGINT	NULL	FK → ProductVariants	Alternative variant if applicable
alternative_notes	NVARCHAR(500)	NULL		Explanation of alternative
requires_procurement	BIT	NOT NULL	0	True if supply requires purchasing from vendor
procurement_lead_days	INT	NULL		Estimated procurement lead time
reviewed_by_user_id	BIGINT	NULL	FK → auth.Users	User who evaluated this line
reviewed_at	DATETIME2	NULL		When the line was evaluated
notes	NVARCHAR(1000)	NULL		Internal line-level notes
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
updated_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	

Indexes:

Index	Columns	Type
IX_SaleInquiryLines_InquiryId	sale_inquiry_id	Clustered-adjacent
UQ_SaleInquiryLines_InqLine	sale_inquiry_id, line_number	Unique
IX_SaleInquiryLines_Product	product_id	Non-unique, filtered (WHERE NOT NULL)
IX_SaleInquiryLines_Rejection	rejection_reason_id	Non-unique, filtered (WHERE NOT NULL)

3.4 Sale Inquiry State Machine



Transition Rules:

From	To	Trigger	Condition
RECEIVED	UNDER_REVIEW	User begins evaluation	At least one line exists
UNDER_REVIEW	REVIEW_COMPLETE	User marks review complete	All lines have a status ≠ PENDING
UNDER_REVIEW	DECLINED	User declines entire inquiry	—
REVIEW_COMPLETE	QUOTED	System — when quotation is created from this inquiry	source_inquiry_id set on quotation
REVIEW_COMPLETE	DECLINED	User declines after review	—

3.5 Line-Level Evaluation Statuses

Status	Meaning	Required Fields
PENDING	Not yet evaluated	—
CAN_SUPPLY	Organization can deliver the requested qty by the date	estimated_delivery_date
PARTIAL	Organization can supply a partial quantity	can_supply_quantity, estimated_delivery_date
CANNOT_SUPPLY	Organization cannot supply this item	rejection_reason_id
UNDER_REVIEW	Evaluation in progress — awaiting internal check (e.g., procurement lead time)	—

3.6 Inquiry Attachments

New Table: `demand.SaleInquiryAttachments`

Column	Type	Nullable	Description
attachment_id	BIGINT IDENTITY	NOT NULL	PK
sale_inquiry_id	BIGINT	NOT NULL	FK → SaleInquiries
org_id	INT	NOT NULL	Tenant discriminator
file_name	NVARCHAR(255)	NOT NULL	Original file name
file_path	NVARCHAR(500)	NOT NULL	Storage path (Azure Blob / local)
file_size_bytes	BIGINT	NOT NULL	
content_type	NVARCHAR(100)	NOT NULL	MIME type
uploaded_by	BIGINT	NOT NULL	FK → auth.Users
uploaded_at	DATETIME2	NOT NULL	SYSUTCDATETIME()

Use Case: Customer sends RFQ as PDF via email. Sales rep creates an inquiry, enters the line items, and attaches the original PDF for reference.

4. C2 — Sale Quotation

4.1 Overview

A Sale Quotation is the organization's formal pricing proposal to a customer. It may originate from a reviewed Sale Inquiry or be created independently. The quotation supports:

- **Line-level rejection with alternatives** — a line can be marked as rejected with a reason, and one or more alternative products offered in its place
- **Per-line customer response tracking** — each line tracks whether the customer accepted, rejected, or countered the offer
- **Validity period** — quotation carries valid_from/valid_to dates
- **Conversion to Sale Order** — accepted lines flow into a Sale Order

 **NOTE:** Sale Quotation is SEPARATE from the existing RFQ (Request for Quotation) implementation in the procurement module. RFQ is buyer-side (sent to suppliers); Sale Quotation is seller-side (sent to customers).

4.2 Schema — SaleQuotations

New Table: `demand.SaleQuotations`

Column	Type	Nullable	Default	Description
sale_quotation_id	BIGINT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
quotation_number	NVARCHAR(20)	NOT NULL	Auto	Pattern: SQ-{YYYY}-{NNNN}, unique per org
customer_id	BIGINT	NOT NULL	FK → BusinessPartners	Customer receiving the quotation
customer_reference	NVARCHAR(50)	NULL		Customer's reference (their RFQ number, inquiry ref, etc.)
customer_reference_date	DATE	NULL		Date on customer's reference document
source_inquiry_id	BIGINT	NULL	FK → SaleInquiries	Originating inquiry (NULL if created independently)
currency_code	NVARCHAR(3)	NOT NULL	FK → Currencies	Quotation currency
valid_from	DATE	NOT NULL	GETDATE()	Quotation validity start
valid_to	DATE	NOT NULL		Quotation validity end
status	NVARCHAR(20)	NOT NULL	'DRAFT'	State machine: DRAFT → SENT → ACCEPTED → REJECTED → EXPIRED → CONVERTED
payment_terms	NVARCHAR(200)	NULL		Payment terms text

Column	Type	Nullable	Default	Description
delivery_terms	NVARCHAR(200)	NULL		Delivery/shipping terms text
subtotal	DECIMAL(18,4)	NOT NULL	0	Sum of line totals (excl. tax)
tax_amount	DECIMAL(18,4)	NOT NULL	0	Total tax
discount_amount	DECIMAL(18,4)	NOT NULL	0	Total discount
grand_total	DECIMAL(18,4)	NOT NULL	0	Final total
notes	NVARCHAR(2000)	NULL		Terms, conditions, or notes printed on quotation
internal_notes	NVARCHAR(2000)	NULL		Internal notes (not sent to customer)
sent_at	DATETIME2	NULL		When quotation was sent to customer
sent_by_user_id	BIGINT	NULL	FK → auth.Users	Who sent it
created_by	BIGINT	NOT NULL	FK → auth.Users	
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
updated_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
trace_id	UNIQUEIDENTIFIER	NOT NULL	NEWID()	

Indexes:

Index	Columns	Type
UQ_SaleQuotations_OrgNumber	org_id, quotation_number	Unique
IX_SaleQuotations_Customer	customer_id	Non-unique
IX_SaleQuotations_Status	org_id, status	Non-unique
IX_SaleQuotations_SourceInquiry	source_inquiry_id	Non-unique, filtered (WHERE NOT NULL)
IX_SaleQuotations_ValidTo	org_id, valid_to	Non-unique (for expiry batch job)

4.3 Schema — SaleQuotationLines

New Table: `demand.SaleQuotationLines`

Column	Type	Nullable	Default	Description
quotation_line_id	BIGINT IDENTITY	NOT NULL	PK	Primary key
sale_quotation_id	BIGINT	NOT NULL	FK → SaleQuotations	Parent quotation
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
line_number	INT	NOT NULL	Auto	Sequential within quotation
source_inquiry_line_id	BIGINT	NULL	FK → SaleInquiryLines	Originating inquiry line (NULL if not from inquiry)
variant_id	BIGINT	NOT NULL	FK → ProductVariants	Product variant being quoted
product_description	NVARCHAR(500)	NOT NULL		Product description (snapshot at quotation time)
quantity	DECIMAL(18,4)	NOT NULL		Offered quantity
uom_id	INT	NOT NULL	FK → UnitOfMeasures	Unit of measure
unit_price	DECIMAL(18,4)	NOT NULL		Price per unit
discount_percent	DECIMAL(5,2)	NOT NULL	0	Line discount %
tax_percent	DECIMAL(5,2)	NOT NULL	0	Tax %
tax_amount	DECIMAL(18,4)	NOT NULL	0	Computed tax amount
line_total	DECIMAL(18,4)	NOT NULL	0	$(qty \times unit_price \times (1 - discount\%)) + tax$

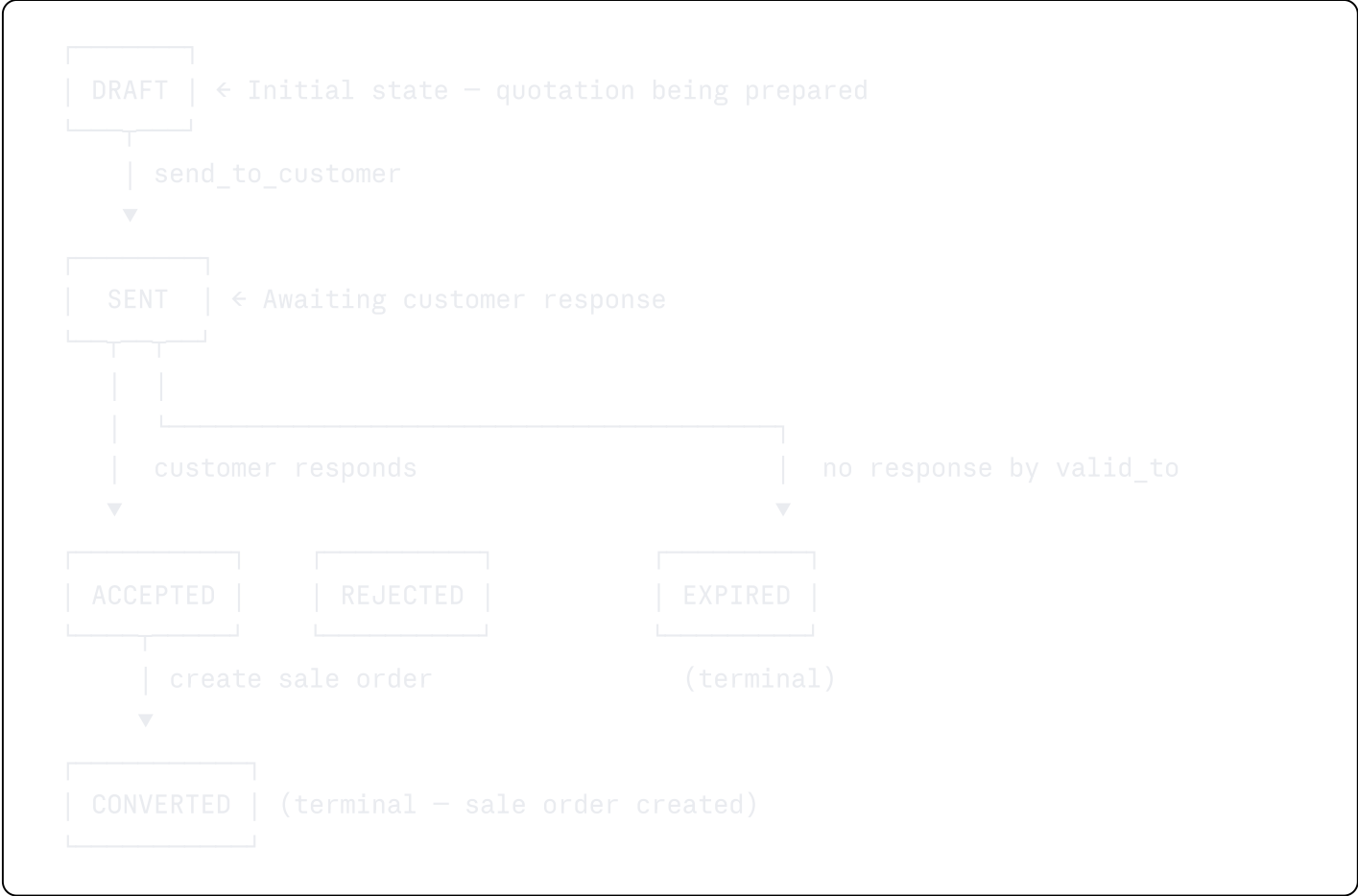
Column	Type	Nullable	Default	Description
promised_delivery_date	DATE	NULL		Promised delivery for this line
line_type	NVARCHAR(15)	NOT NULL	'NORMAL'	NORMAL, ALTERNATIVE, or REJECTED
rejection_reason_id	INT	NULL	FK → RejectionReasons	Why this line is rejected (when line_type = REJECTED)
rejection_notes	NVARCHAR(500)	NULL		Rejection explanation
alternative_for_line_id	BIGINT	NULL	FK → self (quotation_line_id)	Points to the REJECTED line this is an alternative for
alternative_notes	NVARCHAR(500)	NULL		Explanation of the alternative
customer_response	NVARCHAR(15)	NOT NULL	'PENDING'	PENDING, ACCEPTED, REJECTED, COUNTER
customer_response_date	DATE	NULL		When customer responded to this line
customer_response_notes	NVARCHAR(500)	NULL		Customer's notes or feedback
customer_counter_price	DECIMAL(18,4)	NULL		Customer's counter-offer price (when response = COUNTER)
notes	NVARCHAR(1000)	NULL		Internal line notes
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	

Column	Type	Nullable	Default	Description
updated_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	

Indexes:

Index	Columns	Type
IX_SaleQuotationLines_QuotationId	sale_quotation_id	Clustered-adjacent
UQ_SaleQuotationLines_QuotLine	sale_quotation_id, line_number	Unique
IX_SaleQuotationLines_Variant	variant_id	Non-unique
IX_SaleQuotationLines_AltFor	alternative_for_line_id	Non-unique, filtered (WHERE NOT NULL)
IX_SaleQuotationLines_CustResponse	sale_quotation_id, customer_response	Non-unique

4.4 Sale Quotation State Machine



Transition Rules:

From	To	Trigger	Condition
DRAFT	SENT	User sends quotation	At least one NORMAL or ALTERNATIVE line exists
SENT	ACCEPTED	User records customer acceptance	At least one line has customer_response = ACCEPTED
SENT	REJECTED	User records customer rejection	All lines have customer_response = REJECTED
SENT	EXPIRED	System batch job (Hangfire)	SYSUTCDATETIME() > valid_to AND status = SENT
ACCEPTED	CONVERTED	System — when Sale Order is created from this quotation	source_quotation_id set on Sale Order
DRAFT	DRAFT	User edits quotation	Lines can be added/modified/removed while draft

 **CRITICAL:** A quotation in SENT status cannot be edited. To modify a sent quotation, the user must create a new quotation (optionally copying lines from the original). The original can be manually set to REJECTED or allowed to expire.

4.5 Line Type & Alternative Flow

Three line types exist on a quotation:

line_type	Description	alternative_for_line_id
NORMAL	Standard quoted item	NULL
REJECTED	Item the org cannot or will not supply	NULL
ALTERNATIVE	Offered replacement for a REJECTED line	Points to the rejected line's quotation_line_id

Example Flow:

```

Line 1: Product A – 100 units @ $50      line_type=NORMAL
Line 2: Product B – 200 units            line_type=REJECTED (reason: OOS)
Line 3: Product B2 – 200 units @ $48     line_type=ALTERNATIVE
alternative_for_line_id=Line 2
Line 4: Product B3 – 150 units @ $45     line_type=ALTERNATIVE
alternative_for_line_id=Line 2
Line 5: Product C – 50 units @ $120     line_type=NORMAL

```

When the customer responds:

- Line 1: customer_response = ACCEPTED
- Line 2: (skipped — already REJECTED by seller)
- Line 3: customer_response = ACCEPTED (customer chose this alternative)
- Line 4: customer_response = REJECTED (customer did not want this one)
- Line 5: customer_response = COUNTER, customer_counter_price = \$110

Conversion to Sale Order: Only lines with `customer_response = ACCEPTED` flow into the Sale Order. COUNTER lines require the seller to accept the counter-price (by updating the line's unit_price and setting customer_response = ACCEPTED) before conversion.

4.6 Quotation Attachments

New Table: `demand.SaleQuotationAttachments`

Column	Type	Nullable	Description
attachment_id	BIGINT IDENTITY	NOT NULL	PK
sale_quotation_id	BIGINT	NOT NULL	FK → SaleQuotations
org_id	INT	NOT NULL	Tenant discriminator
file_name	NVARCHAR(255)	NOT NULL	Original file name
file_path	NVARCHAR(500)	NOT NULL	Storage path
file_size_bytes	BIGINT	NOT NULL	
content_type	NVARCHAR(100)	NOT NULL	MIME type
uploaded_by	BIGINT	NOT NULL	FK → auth.Users
uploaded_at	DATETIME2	NOT NULL	SYSUTCDATETIME()

5. C3 — Sale Order Extensions (Source Linking & Customer PO Capture)

5.1 Overview

The existing `demand.SaleOrders` table is extended with columns that:

1. **Link back to the source** — which quotation and/or inquiry this Sale Order originated from
2. **Capture the customer's Purchase Order** — the customer's PO number, date, and attached PO document
3. **Record how the order was created** — manual entry, from quotation, portal, or inter-tenant

5.2 Schema Change

Modify: `demand.SaleOrders`

SaleOrders — New Columns (ADD)

Column	Type	Nullable	Default	Description
source_quotation_id	BIGINT	NULL	FK → SaleQuotations	Originating quotation (NULL for direct orders)
source_inquiry_id	BIGINT	NULL	FK → SaleInquiries	Originating inquiry (NULL when no inquiry)
source_type	NVARCHAR(20)	NOT NULL	'MANUAL'	MANUAL, FROM_QUOTATION, PORTAL, INTER_TENANT
customer_po_reference	NVARCHAR(50)	NULL		Customer's Purchase Order number
customer_po_date	DATE	NULL		Date on customer's PO
customer_po_attachment_id	BIGINT	NULL	FK → demand.SaleOrderAttachments	Attached customer PO document

New Indexes:

Index	Columns	Type
IX_SaleOrders_SourceQuotation	source_quotation_id	Non-unique, filtered (WHERE NOT NULL)
IX_SaleOrders_SourceInquiry	source_inquiry_id	Non-unique, filtered (WHERE NOT NULL)
IX_SaleOrders_CustomerPO	org_id, customer_po_reference	Non-unique, filtered (WHERE NOT NULL)

5.3 Sale Order Attachments

New Table: `demand.SaleOrderAttachments`

Column	Type	Nullable	Description
attachment_id	BIGINT IDENTITY	NOT NULL	PK
sale_order_id	BIGINT	NOT NULL	FK → SaleOrders
org_id	INT	NOT NULL	Tenant discriminator
file_name	NVARCHAR(255)	NOT NULL	Original file name
file_path	NVARCHAR(500)	NOT NULL	Storage path
file_size_bytes	BIGINT	NOT NULL	
content_type	NVARCHAR(100)	NOT NULL	MIME type
attachment_type	NVARCHAR(20)	NOT NULL	'CUSTOMER_PO' or 'GENERAL'
uploaded_by	BIGINT	NOT NULL	FK → auth.Users
uploaded_at	DATETIME2	NOT NULL	SYSUTCDATETIME()

5.4 Source Type Enum

Value	Description	Typical Fields Populated
MANUAL	Entered directly by sales rep	customer_po_reference (optional)
FROM_QUOTATION	Converted from an accepted Sale Quotation	source_quotation_id, source_inquiry_id (if chain), customer_po_reference
PORTAL	Customer created via self-service portal (future)	customer_po_reference
INTER_TENANT	Created from another tenant's Purchase Order (future)	customer_po_reference

5.5 Conversion Logic — Quotation to Sale Order

When creating a Sale Order from a Sale Quotation:

```
csharp
```

```

// Pseudocode - Quotation → Sale Order conversion
var quotation = await _quotationRepo.GetWithLinesAsync(quotationId);

// Only lines the customer accepted are converted
var acceptedLines = quotation.Lines
    .Where(l => l.CustomerResponse == "ACCEPTED")
    .ToList();

if (!acceptedLines.Any())
    throw new BusinessException("No accepted lines to convert");

var saleOrder = new SaleOrder
{
    OrgId = quotation.OrgId,
    PartnerId = quotation.CustomerId,
    SourceQuotationId = quotation.SaleQuotationId,
    SourceInquiryId = quotation.SourceInquiryId, // chain through
    SourceType = "FROM_QUOTATION",
    CurrencyCode = quotation.CurrencyCode,
    OrderDate = DateOnly.FromDateTime(DateTime.UtcNow),
    Status = "DRAFT",
    CustomerPoReference = dto.CustomerPoReference, // captured at conversion
    CustomerPoDate = dto.CustomerPoDate
};

foreach (var ql in acceptedLines)
{
    saleOrder.Lines.Add(new SaleOrderLine
    {
        VariantId = ql.VariantId,
        Quantity = ql.Quantity,
        UnitPrice = ql.UnitPrice,
        DiscountPercent = ql.DiscountPercent,
        TaxPercent = ql.TaxPercent,
        Status = "OPEN"
    });
}

await _saleOrderRepo.CreateAsync(saleOrder);

// Transition quotation → CONVERTED
quotation.Status = "CONVERTED";
await _quotationRepo.UpdateAsync(quotation);

```

6. C4 — Sale Order Line Delivery Indicators & Inventory Reservation

6.1 Overview

Two enhancements to Sale Order lines:

1. **Delivery Progress Indicators** — a computed green/yellow/red visual indicator per line showing fulfillment status (similar to Odoo's delivery status tooltips)
2. **Inventory Reservation Column** — a per-line `reserved_qty` and an RBAC-controlled "Reserve" button that calls `IStockReservationService`

6.2 Schema Change — SaleOrderLines

Modify: `demand.SaleOrderLines`

SaleOrderLines — New Column (ADD)

Column	Type	Nullable	Default	Description
reserved_qty	DECIMAL(18,4)	NOT NULL	0	Currently reserved quantity from inventory

 **NOTE:** The delivery indicator is NOT stored — it is computed at read time from `fulfilled_qty`, `reserved_qty`, and `quantity`.

6.3 Delivery Indicator Logic (Computed)

The delivery indicator is a computed property in the DTO, never persisted:

```
csharp
```

```
// Pseudocode — SaleOrderLineDto.DeliveryIndicator
public string DeliveryIndicator => Status switch
{
    "CANCELLED" => "GREY",
    "FULFILLED" or "INVOICED" => "GREEN",
    _ => ComputeIndicator()
};

private string ComputeIndicator()
{
    if (FulfilledQty >= Quantity) return "GREEN"; // fully delivered
    if (FulfilledQty > 0) return "YELLOW"; // partially delivered
    if (ReservedQty >= Quantity) return "BLUE"; // fully reserved, not yet shipped
    if (ReservedQty > 0) return "YELLOW"; // partially reserved
    return "RED"; // nothing reserved or delivered
}
```

Indicator	Color	Meaning
GREEN		Fully fulfilled — entire quantity delivered
BLUE		Fully reserved — inventory held, not yet shipped
YELLOW		Partially fulfilled or partially reserved
RED		Nothing reserved or fulfilled — action needed
GREY		Cancelled line

6.4 Inventory Reservation Flow

Reserve inventory for a Sale Order line:

User clicks [Reserve] on SO line

- System checks: user has 'sales_reserve_inventory' permission?
- NO: button is disabled (greyed out), tooltip: "Insufficient permissions"
- YES:
 - Check available_qty for variant in warehouse
 - available_qty >= remaining unreserved qty?
 - YES: call IStockReservationService.Reserve(
variantId, warehouseId, reserveQty,
sourceType = SALES_ORDER,
sourceId = saleOrderId)
Update SaleOrderLine.reserved_qty += reserveQty
Indicator may change: RED→BLUE or YELLOW→BLUE
 - NO: show dialog "Available: {X}, Required: {Y}. Reserve partial?"
 - User confirms partial → reserve available
 - User cancels → no action

Release reserved inventory:

User clicks [Release] on SO line (visible when reserved_qty > 0)

- System checks: user has 'sales_reserve_inventory' permission?
- YES:
 - IStockReservationService.Release(reservationId)
 - SaleOrderLine.reserved_qty -= releasedQty
 - stock_balance.reserved_qty decreased
 - stock_balance.available_qty increased
 - Indicator changes: BLUE→RED or YELLOW→RED

Automatic release on Sale Order cancellation:

When SaleOrder.Status → CANCELLED:

- For each line with reserved_qty > 0:
 - IStockReservationService.Release(reservationId)
 - reserved_qty set to 0
 - Inventory returned to available pool

6.5 Reservation Permissions

New Permission Claims:

Claim	Description	Default Roles
sales_reserve_inventory	Reserve inventory against a Sale Order line	SALES_MANAGER, WAREHOUSE_MANAGER
sales_release_reservation	Release existing reservation on a Sale Order line	SALES_MANAGER, WAREHOUSE_MANAGER

UI Behavior:

User Has Permission	Reserve Button	Release Button
Neither	Disabled (greyed)	Disabled (greyed)
sales_reserve_inventory only	Enabled	Disabled
sales_release_reservation only	Disabled	Enabled
Both	Enabled	Enabled

⚠ CRITICAL: The Reserve/Release buttons are DISABLED in the UI when the user lacks the required permission. They are not hidden — hiding them would confuse users who expect to see them. The disabled state includes a tooltip explaining why: "You do not have permission to reserve inventory. Contact your administrator."

6.6 Reservation TTL

The existing `demand.SaleOrderConfig.reservation_ttl_hours` applies. When a reservation exceeds the TTL without the Sale Order advancing to fulfillment, the Hangfire `ReservationExpiryJob` releases it:

```
csharp
// Existing behavior – no changes needed
// ReservationExpiryJob runs on schedule, checks:
//   reservation.created_at + config.reservation_ttl_hours < SYSUTCDATETIME()
//   AND reservation.status == ACTIVE
//   → release reservation, update SO line reserved_qty
```

7. C5 — Rejection Reason Lookup

7.1 Overview

A shared lookup table for rejection reasons used by both Sale Inquiry lines and Sale Quotation lines. The rejection reasons enable analytics: "Why are we declining customer requests?"

7.2 Schema — RejectionReasons

New Table: `lookups.RejectionReasons`

Column	Type	Nullable	Default	Description
rejection_reason_id	INT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
code	NVARCHAR(10)	NOT NULL		Short code (e.g., OOS, MOQ, DIS)
description	NVARCHAR(200)	NOT NULL		Human-readable description
is_active	BIT	NOT NULL	1	Soft delete
display_order	INT	NOT NULL	0	Sort order in dropdowns
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
updated_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	

Indexes:

Index	Columns	Type
UQ_RejectionReasons_OrgCode	org_id, code	Unique

7.3 Seed Data

Code	Description
NIP	Not in product catalog
OOS	Out of stock — no replenishment planned
DIS	Product discontinued
MOQ	Below minimum order quantity
GEO	Cannot deliver to requested region
REG	Regulatory restriction on this product
CAP	Insufficient production capacity
CRD	Customer credit hold
PRC	Cannot meet requested price
OTH	Other (see notes)



NOTE: Seed data is inserted per org on tenant provisioning. Organizations can add, rename, or deactivate reasons but cannot delete seed codes.

8. Database Migrations

Seven migrations are required for this addendum. They are additive and non-breaking.

8.1 Migration Order

#	Migration Name	Schema	Type	Description
M1	CreateRejectionReasons	lookups	CREATE TABLE	New lookup table for rejection reasons + seed data
M2	CreateSaleInquiries	demand	CREATE TABLE	SaleInquiries + SaleInquiryLines + SaleInquiryAttachments
M3	CreateSaleQuotations	demand	CREATE TABLE	SaleQuotations + SaleQuotationLines + SaleQuotationAttachments
M4	ExtendSaleOrdersSourceLinking	demand	ALTER TABLE	Add source columns + customer PO to SaleOrders
M5	CreateSaleOrderAttachments	demand	CREATE TABLE	SaleOrderAttachments table
M6	AddReservedQtyToSaleOrderLines	demand	ALTER TABLE	Add reserved_qty to SaleOrderLines
M7	SeedSalesPermissions	auth	INSERT	Add sales_reserve_inventory and sales_release_reservation claims

8.2 Migration Details

M1 — CreateRejectionReasons

sql

```

CREATE TABLE lookups.RejectionReasons (
    rejection_reason_id INT IDENTITY(1,1) NOT NULL
        CONSTRAINT PK_RejectionReasons PRIMARY KEY,
    org_id INT NOT NULL
        CONSTRAINT FK_RejectionReasons_Org FOREIGN KEY REFERENCES tenants.Organization
        code NVARCHAR(10) NOT NULL,
    description NVARCHAR(200) NOT NULL,
    is_active BIT NOT NULL CONSTRAINT DF_RejectionReasons_IsActive DEFAULT 1,
    display_order INT NOT NULL CONSTRAINT DF_RejectionReasons_DisplayOrder DEFAULT 0,
    created_at DATETIME2 NOT NULL CONSTRAINT DF_RejectionReasons_Created DEFAULT SYSUT
    updated_at DATETIME2 NOT NULL CONSTRAINT DF_RejectionReasons_Updated DEFAULT SYSUT

    CONSTRAINT UQ_RejectionReasons_OrgCode UNIQUE (org_id, code)
);

-- Seed data is inserted per org via tenant provisioning stored procedure
-- See §7.3 for seed values

```

M2 — CreateSaleInquiries

sql

```

CREATE TABLE demand.SaleInquiries (
    sale_inquiry_id BIGINT IDENTITY(1,1) NOT NULL
        CONSTRAINT PK_SaleInquiries PRIMARY KEY,
    org_id INT NOT NULL
        CONSTRAINT FK_SaleInquiries_Org FOREIGN KEY REFERENCES tenants.Organizations(
inquiry_number NVARCHAR(20) NOT NULL,
customer_id BIGINT NOT NULL
        CONSTRAINT FK_SaleInquiries_Customer FOREIGN KEY REFERENCES suppliers.Business
customer_reference NVARCHAR(50) NULL,
customer_reference_date DATE NULL,
status NVARCHAR(20) NOT NULL CONSTRAINT DF_SaleInquiries_Status DEFAULT 'RECEIVED
received_date DATE NOT NULL CONSTRAINT DF_SaleInquiries_ReceivedDate DEFAULT GETDA
response_deadline DATE NULL,
assigned_to_user_id BIGINT NULL,
notes NVARCHAR(2000) NULL,
created_by BIGINT NOT NULL,
created_at DATETIME2 NOT NULL CONSTRAINT DF_SaleInquiries_Created DEFAULT SYSUTCD
updated_at DATETIME2 NOT NULL CONSTRAINT DF_SaleInquiries_Updated DEFAULT SYSUTCD
trace_id UNIQUEIDENTIFIER NOT NULL CONSTRAINT DF_SaleInquiries_TraceId DEFAULT NEW

    CONSTRAINT UQ_SaleInquiries_OrgNumber UNIQUE (org_id, inquiry_number)
);

```

```

CREATE INDEX IX_SaleInquiries_Customer ON demand.SaleInquiries(customer_id);
CREATE INDEX IX_SaleInquiries_Status ON demand.SaleInquiries(org_id, status);
CREATE INDEX IX_SaleInquiries_AssignedTo ON demand.SaleInquiries(assigned_to_user_id)
    WHERE assigned_to_user_id IS NOT NULL;

```

```

CREATE TABLE demand.SaleInquiryLines (
    inquiry_line_id BIGINT IDENTITY(1,1) NOT NULL
        CONSTRAINT PK_SaleInquiryLines PRIMARY KEY,
    sale_inquiry_id BIGINT NOT NULL
        CONSTRAINT FK_SaleInquiryLines_Inquiry FOREIGN KEY REFERENCES demand.SaleInqu
    org_id INT NOT NULL
        CONSTRAINT FK_SaleInquiryLines_Org FOREIGN KEY REFERENCES tenants.Organization
    line_number INT NOT NULL,
    product_id BIGINT NULL,
    variant_id BIGINT NULL,
    product_description NVARCHAR(500) NOT NULL,
    requested_quantity DECIMAL(18,4) NOT NULL,
    requested_uom_id INT NULL,
    requested_delivery_date DATE NULL,
    line_status NVARCHAR(20) NOT NULL CONSTRAINT DF_SaleInquiryLines_Status DEFAULT '

```

```

can_supply_quantity DECIMAL(18,4) NULL,
estimated_delivery_date DATE NULL,
rejection_reason_id INT NULL
    CONSTRAINT FK_SaleInquiryLines_RejReason FOREIGN KEY REFERENCES lookups.RejectionReasons (rejection_reason_id)
rejection_notes NVARCHAR(500) NULL,
alternative_product_id BIGINT NULL,
alternative_variant_id BIGINT NULL,
alternative_notes NVARCHAR(500) NULL,
requires_procurement BIT NOT NULL CONSTRAINT DF_SaleInquiryLines_ReqProc DEFAULT 0
procurement_lead_days INT NULL,
reviewed_by_user_id BIGINT NULL,
reviewed_at DATETIME2 NULL,
notes NVARCHAR(1000) NULL,
created_at DATETIME2 NOT NULL CONSTRAINT DF_SaleInquiryLines_Created DEFAULT SYSUT
updated_at DATETIME2 NOT NULL CONSTRAINT DF_SaleInquiryLines_Updated DEFAULT SYSUT

CONSTRAINT UQ_SaleInquiryLines_InqLine UNIQUE (sale_inquiry_id, line_number)
);

CREATE INDEX IX_SaleInquiryLines_Product ON demand.SaleInquiryLines(product_id)
    WHERE product_id IS NOT NULL;
CREATE INDEX IX_SaleInquiryLines_Rejection ON demand.SaleInquiryLines(rejection_reason_id)
    WHERE rejection_reason_id IS NOT NULL;

CREATE TABLE demand.SaleInquiryAttachments (
    attachment_id BIGINT IDENTITY(1,1) NOT NULL
        CONSTRAINT PK_SaleInquiryAttachments PRIMARY KEY,
    sale_inquiry_id BIGINT NOT NULL
        CONSTRAINT FK_SaleInqAttach_Inquiry FOREIGN KEY REFERENCES demand.SaleInquiryLines (sale_inquiry_id),
    org_id INT NOT NULL,
    file_name NVARCHAR(255) NOT NULL,
    file_path NVARCHAR(500) NOT NULL,
    file_size_bytes BIGINT NOT NULL,
    content_type NVARCHAR(100) NOT NULL,
    uploaded_by BIGINT NOT NULL,
    uploaded_at DATETIME2 NOT NULL CONSTRAINT DF_SaleInqAttach_Uploaded DEFAULT SYSUT
);

```

M3 — CreateSaleQuotations

sql

```

CREATE TABLE demand.SaleQuotations (
    sale_quotation_id BIGINT IDENTITY(1,1) NOT NULL
        CONSTRAINT PK_SaleQuotations PRIMARY KEY,
    org_id INT NOT NULL
        CONSTRAINT FK_SaleQuotations_Org FOREIGN KEY REFERENCES tenants.Organizations,
    quotation_number NVARCHAR(20) NOT NULL,
    customer_id BIGINT NOT NULL
        CONSTRAINT FK_SaleQuotations_Customer FOREIGN KEY REFERENCES suppliers.Business,
    customer_reference NVARCHAR(50) NULL,
    customer_reference_date DATE NULL,
    source_inquiry_id BIGINT NULL
        CONSTRAINT FK_SaleQuotations_Inquiry FOREIGN KEY REFERENCES demand.SaleInquiries,
    currency_code NVARCHAR(3) NOT NULL,
    valid_from DATE NOT NULL CONSTRAINT DF_SaleQuotations_ValidFrom DEFAULT GETDATE(),
    valid_to DATE NOT NULL,
    status NVARCHAR(20) NOT NULL CONSTRAINT DF_SaleQuotations_Status DEFAULT 'DRAFT',
    payment_terms NVARCHAR(200) NULL,
    delivery_terms NVARCHAR(200) NULL,
    subtotal DECIMAL(18,4) NOT NULL CONSTRAINT DF_SaleQuotations_Subtotal DEFAULT 0,
    tax_amount DECIMAL(18,4) NOT NULL CONSTRAINT DF_SaleQuotations_Tax DEFAULT 0,
    discount_amount DECIMAL(18,4) NOT NULL CONSTRAINT DF_SaleQuotations_Discount DEFAULT 0,
    grand_total DECIMAL(18,4) NOT NULL CONSTRAINT DF_SaleQuotations_Total DEFAULT 0,
    notes NVARCHAR(2000) NULL,
    internal_notes NVARCHAR(2000) NULL,
    sent_at DATETIME2 NULL,
    sent_by_user_id BIGINT NULL,
    created_by BIGINT NOT NULL,
    created_at DATETIME2 NOT NULL CONSTRAINT DF_SaleQuotations_Created DEFAULT SYSUTCTIME(),
    updated_at DATETIME2 NOT NULL CONSTRAINT DF_SaleQuotations_Updated DEFAULT SYSUTCTIME(),
    trace_id UNIQUEIDENTIFIER NOT NULL CONSTRAINT DF_SaleQuotations_TraceId DEFAULT NEWID(),

    CONSTRAINT UQ_SaleQuotations_OrgNumber UNIQUE (org_id, quotation_number)
);

```

```

CREATE INDEX IX_SaleQuotations_Customer ON demand.SaleQuotations(customer_id);
CREATE INDEX IX_SaleQuotations_Status ON demand.SaleQuotations(org_id, status);
CREATE INDEX IX_SaleQuotations_SourceInquiry ON demand.SaleQuotations(source_inquiry_id)
    WHERE source_inquiry_id IS NOT NULL;
CREATE INDEX IX_SaleQuotations_ValidTo ON demand.SaleQuotations(org_id, valid_to);

```

```

CREATE TABLE demand.SaleQuotationLines (
    quotation_line_id BIGINT IDENTITY(1,1) NOT NULL
        CONSTRAINT PK_SaleQuotationLines PRIMARY KEY,

```

```

sale_quotation_id BIGINT NOT NULL
    CONSTRAINT FK_SaleQuotLines_Quotation FOREIGN KEY REFERENCES demand.SaleQuotation(sale_quotation_id),
org_id INT NOT NULL
    CONSTRAINT FK_SaleQuotLines_Org FOREIGN KEY REFERENCES tenants.Organizations(org_id),
line_number INT NOT NULL,
source_inquiry_line_id BIGINT NULL
    CONSTRAINT FK_SaleQuotLines_InqLine FOREIGN KEY REFERENCES demand.SaleInquiryLines(source_inquiry_line_id),
variant_id BIGINT NOT NULL,
product_description NVARCHAR(500) NOT NULL,
quantity DECIMAL(18,4) NOT NULL,
uom_id INT NOT NULL,
unit_price DECIMAL(18,4) NOT NULL,
discount_percent DECIMAL(5,2) NOT NULL CONSTRAINT DF_SaleQuotLines_Discount DEFAULT 0,
tax_percent DECIMAL(5,2) NOT NULL CONSTRAINT DF_SaleQuotLines_TaxPct DEFAULT 0,
tax_amount DECIMAL(18,4) NOT NULL CONSTRAINT DF_SaleQuotLines_TaxAmt DEFAULT 0,
line_total DECIMAL(18,4) NOT NULL CONSTRAINT DF_SaleQuotLines_Total DEFAULT 0,
promised_delivery_date DATE NULL,
line_type NVARCHAR(15) NOT NULL CONSTRAINT DF_SaleQuotLines_LineType DEFAULT 'NORMAL',
rejection_reason_id INT NULL
    CONSTRAINT FK_SaleQuotLines_RejReason FOREIGN KEY REFERENCES lookups.RejectionReasons(rejection_reason_id),
rejection_notes NVARCHAR(500) NULL,
alternative_for_line_id BIGINT NULL
    CONSTRAINT FK_SaleQuotLines_AltFor FOREIGN KEY REFERENCES demand.SaleQuotationLines(alternative_for_line_id),
alternative_notes NVARCHAR(500) NULL,
customer_response NVARCHAR(15) NOT NULL CONSTRAINT DF_SaleQuotLines_CustResp DEFAULT 'N',
customer_response_date DATE NULL,
customer_response_notes NVARCHAR(500) NULL,
customer_counter_price DECIMAL(18,4) NULL,
notes NVARCHAR(1000) NULL,
created_at DATETIME2 NOT NULL CONSTRAINT DF_SaleQuotLines_Created DEFAULT SYSUTCDATETIME(),
updated_at DATETIME2 NOT NULL CONSTRAINT DF_SaleQuotLines_Updated DEFAULT SYSUTCDATETIME(),

CONSTRAINT UQ_SaleQuotationLines_QuotLine UNIQUE (sale_quotation_id, line_number)
);

CREATE INDEX IX_SaleQuotLines_Variant ON demand.SaleQuotationLines(variant_id);
CREATE INDEX IX_SaleQuotLines_AltFor ON demand.SaleQuotationLines(alternative_for_line_id)
    WHERE alternative_for_line_id IS NOT NULL;
CREATE INDEX IX_SaleQuotLines_CustResponse ON demand.SaleQuotationLines(sale_quotation_id, customer_response);

CREATE TABLE demand.SaleQuotationAttachments (
    attachment_id BIGINT IDENTITY(1,1) NOT NULL
        CONSTRAINT PK_SaleQuotationAttachments PRIMARY KEY,
    sale_quotation_id BIGINT NOT NULL

```

```
        CONSTRAINT FK_SaleQuotAttach_Quotation FOREIGN KEY REFERENCES demand.SaleQuotAttach (
org_id INT NOT NULL,
file_name NVARCHAR(255) NOT NULL,
file_path NVARCHAR(500) NOT NULL,
file_size_bytes BIGINT NOT NULL,
content_type NVARCHAR(100) NOT NULL,
uploaded_by BIGINT NOT NULL,
uploaded_at DATETIME2 NOT NULL CONSTRAINT DF_SaleQuotAttach_Uploaded DEFAULT SYSUT
);
```

M4 — ExtendSaleOrdersSourceLinking

sql

```

ALTER TABLE demand.SaleOrders
ADD source_quotation_id BIGINT NULL
    CONSTRAINT FK_SaleOrders_SourceQuotation
    FOREIGN KEY REFERENCES demand.SaleQuotations(sale_quotation_id);

ALTER TABLE demand.SaleOrders
ADD source_inquiry_id BIGINT NULL
    CONSTRAINT FK_SaleOrders_SourceInquiry
    FOREIGN KEY REFERENCES demand.SaleInquiries(sale_inquiry_id);

ALTER TABLE demand.SaleOrders
ADD source_type NVARCHAR(20) NOT NULL
    CONSTRAINT DF_SaleOrders_SourceType DEFAULT 'MANUAL';

ALTER TABLE demand.SaleOrders
ADD customer_po_reference NVARCHAR(50) NULL;

ALTER TABLE demand.SaleOrders
ADD customer_po_date DATE NULL;

ALTER TABLE demand.SaleOrders
ADD customer_po_attachment_id BIGINT NULL;

-- Indexes
CREATE INDEX IX_SaleOrders_SourceQuotation ON demand.SaleOrders(source_quotation_id)
    WHERE source_quotation_id IS NOT NULL;

CREATE INDEX IX_SaleOrders_SourceInquiry ON demand.SaleOrders(source_inquiry_id)
    WHERE source_inquiry_id IS NOT NULL;

CREATE INDEX IX_SaleOrders_CustomerPO ON demand.SaleOrders(org_id, customer_po_reference)
    WHERE customer_po_reference IS NOT NULL;

-- Backfill: all existing Sale Orders get source_type = 'MANUAL'
-- (handled by DEFAULT constraint – no data migration needed)

```

M5 — CreateSaleOrderAttachments

sql

```

CREATE TABLE demand.SaleOrderAttachments (
    attachment_id BIGINT IDENTITY(1,1) NOT NULL
        CONSTRAINT PK_SaleOrderAttachments PRIMARY KEY,
    sale_order_id BIGINT NOT NULL
        CONSTRAINT FK_SOAttach_SaleOrder FOREIGN KEY REFERENCES demand.SaleOrders(sale_order_id),
    org_id INT NOT NULL,
    file_name NVARCHAR(255) NOT NULL,
    file_path NVARCHAR(500) NOT NULL,
    file_size_bytes BIGINT NOT NULL,
    content_type NVARCHAR(100) NOT NULL,
    attachment_type NVARCHAR(20) NOT NULL CONSTRAINT DF_SOAttach_Type DEFAULT 'GENERAL',
    uploaded_by BIGINT NOT NULL,
    uploaded_at DATETIME2 NOT NULL CONSTRAINT DF_SOAttach_Uploaded DEFAULT SYSUTCDATE()
);

CREATE INDEX IX_SOAttach_SaleOrder ON demand.SaleOrderAttachments(sale_order_id);

```

M6 — AddReservedQtyToSaleOrderLines

sql

```

ALTER TABLE demand.SaleOrderLines
ADD reserved_qty DECIMAL(18,4) NOT NULL
    CONSTRAINT DF_SaleOrderLines_ReservedQty DEFAULT 0;

-- Backfill: existing lines with status='RESERVED' should have
-- reserved_qty populated from StockReservations
UPDATE sol
SET sol.reserved_qty = ISNULL(sr.total_reserved, 0)
FROM demand.SaleOrderLines sol
OUTER APPLY (
    SELECT SUM(r.quantity) AS total_reserved
    FROM inventory.StockReservations r
    WHERE r.source_type = 'SALES_ORDER'
        AND r.source_id = sol.sale_order_id
        AND r.variant_id = sol.variant_id
        AND r.status = 'ACTIVE'
) sr
WHERE sol.status = 'RESERVED';

```

M7 — SeedSalesPermissions

```
sql

-- Insert new permission claims for sales reservation
INSERT INTO auth.Permissions (permission_code, permission_name, module, description)
VALUES
    ('sales_reserve_inventory', 'Reserve Inventory for Sale Order', 'Sales',
    'Allows reserving inventory against Sale Order lines'),
    ('sales_release_reservation', 'Release Sale Order Reservation', 'Sales',
    'Allows releasing reserved inventory from Sale Order lines');

-- Assign to default roles (org-specific role assignment via tenant provisioning)
-- SALES_MANAGER and WAREHOUSE_MANAGER roles get both permissions
```

8.3 Rollback

Migration Rollback Steps

M1	DROP TABLE lookups.RejectionReasons (after clearing FKs from M2, M3)
M2	DROP TABLE demand.SaleInquiryAttachments, demand.SaleInquiryLines, demand.SaleInquiries
M3	DROP TABLE demand.SaleQuotationAttachments, demand.SaleQuotationLines, demand.SaleQuotations
M4	DROP COLUMN source_quotation_id, source_inquiry_id, source_type, customer_po_reference, customer_po_date, customer_po_attachment_id from SaleOrders (after clearing FKs and indexes)
M5	DROP TABLE demand.SaleOrderAttachments
M6	DROP COLUMN reserved_qty from SaleOrderLines
M7	DELETE inserted permission claims

9. API Changes

9.1 New Endpoints — Sale Inquiry

Endpoint	Method	Description	Permission
/api/sale-inquiries	GET	List inquiries (paginated, filterable by status, customer, date range)	sales_inquiry_view
/api/sale-inquiries	POST	Create a new inquiry	sales_inquiry_create
/api/sale-inquiries/{id}	GET	Get inquiry with lines	sales_inquiry_view
/api/sale-inquiries/{id}	PUT	Update inquiry header	sales_inquiry_edit
/api/sale-inquiries/{id}/lines	POST	Add a line to inquiry	sales_inquiry_edit
/api/sale-inquiries/{id}/lines/{lineId}	PUT	Update a line (including evaluation)	sales_inquiry_edit
/api/sale-inquiries/{id}/lines/{lineId}	DELETE	Remove a line	sales_inquiry_edit
/api/sale-inquiries/{id}/status	PATCH	Transition inquiry status	sales_inquiry_edit
/api/sale-inquiries/{id}/attachments	POST	Upload attachment	sales_inquiry_edit
/api/sale-inquiries/{id}/attachments/{attachId}	DELETE	Remove attachment	sales_inquiry_edit
/api/sale-inquiries/{id}/create-quotation	POST	Create quotation from reviewed inquiry	sales_quotation_create

9.2 New Endpoints — Sale Quotation

Endpoint	Method	Description	Permission
/api/sale-quotations	GET	List quotations (paginated, filterable)	sales_quotation_view
/api/sale-quotations	POST	Create a new quotation	sales_quotation_create
/api/sale-quotations/{id}	GET	Get quotation with lines	sales_quotation_view
/api/sale-quotations/{id}	PUT	Update quotation header (DRAFT only)	sales_quotation_edit
/api/sale-quotations/{id}/lines	POST	Add a line (DRAFT only)	sales_quotation_edit
/api/sale-quotations/{id}/lines/{lineId}	PUT	Update a line (DRAFT only)	sales_quotation_edit
/api/sale-quotations/{id}/lines/{lineId}	DELETE	Remove a line (DRAFT only)	sales_quotation_edit
/api/sale-quotations/{id}/send	POST	Send quotation to customer (DRAFT → SENT)	sales_quotation_send
/api/sale-quotations/{id}/lines/{lineId}/customer-response	PATCH	Record customer response on a line	sales_quotation_edit
/api/sale-quotations/{id}/accept	POST	Mark quotation as accepted	sales_quotation_edit
/api/sale-quotations/{id}/reject	POST	Mark quotation as rejected	sales_quotation_edit
/api/sale-quotations/{id}/convert-to-order	POST	Create Sale Order from accepted quotation	sales_order_create
/api/sale-quotations/{id}/attachments	POST	Upload attachment	sales_quotation_edit

9.3 Modified Endpoints — Sale Order

Endpoint	Change	Change ID
POST /api/sale-orders	Accept source_quotation_id, source_inquiry_id, source_type, customer_po_reference, customer_po_date in request DTO	C3
GET /api/sale-orders/{id}	Return source_quotation_id, source_inquiry_id, source_type, customer_po_reference, customer_po_date, delivery indicators per line, reserved_qty per line	C3, C4
GET /api/sale-orders/{id}/lines	Include reserved_qty and computed delivery_indicator per line	C4

9.4 New Endpoints — Reservation

Endpoint	Method	Description	Permission
/api/sale-orders/{id}/lines/{lineId}/reserve	POST	Reserve inventory for a SO line	sales_reserve_inventory
/api/sale-orders/{id}/lines/{lineId}/release	POST	Release reservation on a SO line	sales_release_reservation
/api/sale-orders/{id}/reserve-all	POST	Batch reserve all open lines	sales_reserve_inventory

9.5 New Endpoints — Rejection Reasons

Endpoint	Method	Description	Permission
/api/rejection-reasons	GET	List active rejection reasons	authenticated
/api/rejection-reasons	POST	Create a new rejection reason	admin
/api/rejection-reasons/{id}	PUT	Update a rejection reason	admin
/api/rejection-reasons/{id}/deactivate	PATCH	Soft-delete a rejection reason	admin

10. UI Wireframes & Specifications

10.1 Sale Inquiry List

Sales → Inquiries

[+ New Inquiry]

Filter: [Status ▼] [Customer ▼] [Date From] [Date To]

Inquiry #	Customer	Received	Status	Lines
INQ-2026-0042	GlobalTech Co	Oct 01	● Review	5
INQ-2026-0041	Meridian Ltd	Sep 28	● Complete	3
INQ-2026-0040	NorthStar Inc	Sep 25	● Quoted	8
INQ-2026-0039	Atlas Corp	Sep 22	● Declined	2

Status legend: ● RECEIVED (blue) ● UNDER_REVIEW (orange)
● REVIEW_COMPLETE (green) ● QUOTED (purple) ● DECLINED (red)

10.2 Sale Inquiry Detail — Line Evaluation

Inquiry: INQ-2026-0042

Customer: GlobalTech Co

Received: Oct 01, 2026

Assigned To: [John Smith ▼]

Status: UNDER_REVIEW

Ref: GT-RFQ-2026-118

Deadline: Oct 10, 2026

[Details] [Lines] [Attachments]

▲ active

#	Product	Qty	UOM	Delivery	Status	Action
1	Steel Rod 10mm	500	KG	Oct 15	● Can	[Edit]
2	Copper Wire 2mm	1000	M	Oct 20	● Partial	[Edit]
	↳ Can supply: 600M by Oct 25					
3	Titanium Sheet	200	KG	Oct 12	● Cannot	[Edit]
	↳ Reason: DIS — Discontinued					
	↳ Alternative: Stainless Steel Sheet 316L					
4	Rubber Gasket	5000	EA	Oct 18	● Review	[Edit]
	↳ Checking supplier lead time...					
5	Bolt M8x40	10000	EA	Oct 10	● Can	[Edit]
	[+ Add Line]					

Status: ● CAN_SUPPLY (green) ● PARTIAL (yellow)

● CANNOT_SUPPLY (red) ● UNDER_REVIEW (orange)

● PENDING (grey)

[Mark Review Complete]

[Decline Inquiry]

10.3 Sale Quotation Detail — Lines with Alternatives

Quotation: SQ-2026-0015

Customer: GlobalTech Co

From Inquiry: INQ-2026-0042

Status: DRAFT

Valid: Oct 03 – Oct 31, 2026

Currency: USD

[Details] [Lines] [Terms] [Attachments]

▲ active

#	Product	Qty	Price	Total	Type	Response
1	Steel Rod 10mm	500	\$2.50	\$1,250	NORMAL	Pending
2	Copper Wire 2mm	600	\$1.80	\$1,080	NORMAL	Pending
	↳ (adjusted from 1000 – partial supply)					
3	Titanium Sheet	—	—	—	✕ REJ	—
	↳ Reason: Discontinued					
3a	SS Sheet 316L	200	\$8.50	\$1,700	◇ ALT	Pending
	↳ Alternative for line 3					
3b	SS Sheet 304	200	\$6.20	\$1,240	◇ ALT	Pending
	↳ Alternative for line 3 (lower grade option)					
4	Bolt M8x40	10000	\$0.12	\$1,200	NORMAL	Pending
	[+ Add Line]					

Subtotal: \$6,470.00

Tax: \$0.00

Grand Total: \$6,470.00

[Send to Customer]

10.4 Sale Quotation — Customer Response Recording

Quotation: SQ-2026-0015

Status: SENT

Record Customer Response

#	Product	Qty	Price	Response
1	Steel Rod 10mm	500	\$2.50	[Accepted ▼]
2	Copper Wire 2mm	600	\$1.80	[Accepted ▼]
3a	SS Sheet 316L	200	\$8.50	[Accepted ▼]
3b	SS Sheet 304	200	\$6.20	[Rejected ▼]
4	Bolt M8x40	10000	\$0.12	[Counter ▼]
↳ Counter price: [\$0.10]				
↳ Notes: [Bulk discount expected for 10K+ qty]				

[Save Responses]

[Mark Quotation Accepted]

10.5 Sale Order — Extended Header with Source & Customer PO

Sale Order: SO-1085

Status: CONFIRMED

Customer: GlobalTech Co

Source: From Quotation SQ-2026-0015 ← clickable link

Inquiry: INQ-2026-0042 ← clickable link

Customer PO

PO Reference	PO Date
GT-PO-2026-4521	Oct 05, 2026

 GlobalTech_PO_4521.pdf [View] [Replace]

[Header]

[Lines]

[Fulfillment]

[Attachments]

[History]

10.6 Sale Order Lines — Delivery Indicators & Reservation

[Header] [Lines] [Fulfillment] [Attachments] [History]

▲ active

	Product	Qty	Rsvd	Dlvr	Status	Ind	Action
1	Steel Rod 10mm	500	500	0	RSVD		[Rlse]
2	Copper Wire 2mm	600	0	300	PARTIAL		[Rsrv]
3	SS Sheet 316L	200	0	200	FULFL		—
4	Bolt M8x40	10000	0	0	OPEN		[Rsrv]

Column Key:

Qty = ordered quantity

Rsvd = reserved_qty (inventory held)

Dlvr = fulfilled_qty (shipped/delivered)

Ind = delivery indicator (computed)

[Reserve All Open Lines]

ⓘ Reserve/Release buttons require 'sales_reserve_inventory' permission. Contact your administrator if disabled.

11. Business Rules

Consolidated business rules for all 5 changes.

Rule ID	Change	Rule Description	Enforcement Point
BR-C1-01	C1	Inquiry can only be created for a BusinessPartner with partner_type including CUSTOMER	SaleInquiryService
BR-C1-02	C1	Inquiry number auto-generated: INQ-{YYYY}-{NNNN} per org per year	SaleInquiryService
BR-C1-03	C1	UNDER_REVIEW → REVIEW_COMPLETE requires ALL lines to have status ≠ PENDING	SaleInquiryService
BR-C1-04	C1	Line status CANNOT_SUPPLY requires rejection_reason_id to be set	SaleInquiryLineService
BR-C1-05	C1	Line status PARTIAL requires can_supply_quantity > 0 and < requested_quantity	SaleInquiryLineService
BR-C1-06	C1	Inquiry in QUOTED or DECLINED status is read-only — no line modifications	SaleInquiryService
BR-C1-07	C1	Creating a quotation from an inquiry sets inquiry status to QUOTED	SaleQuotationService
BR-C2-01	C2	Quotation can only be created for a BusinessPartner with partner_type including CUSTOMER	SaleQuotationService
BR-C2-02	C2	Quotation number auto-generated: SQ-{YYYY}-{NNNN} per org per year	SaleQuotationService
BR-C2-03	C2	valid_to must be ≥ valid_from	SaleQuotationService
BR-C2-04	C2	DRAFT → SENT requires at least one line with line_type NORMAL or ALTERNATIVE	SaleQuotationService
BR-C2-05	C2	Lines can only be added/modified/removed while quotation is DRAFT	SaleQuotationLineService
BR-C2-06	C2	line_type = ALTERNATIVE requires alternative_for_line_id pointing to a REJECTED line in the same quotation	SaleQuotationLineService
BR-C2-07	C2	SENT → ACCEPTED requires at least one line with customer_response = ACCEPTED	SaleQuotationService

Rule ID	Change	Rule Description	Enforcement Point
BR-C2-08	C2	SENT → REJECTED: all non-REJECTED lines must have customer_response = REJECTED	SaleQuotationService
BR-C2-09	C2	customer_response = COUNTER requires customer_counter_price > 0	SaleQuotationLineService
BR-C2-10	C2	Quotation expiry job runs daily — marks SENT quotations past valid_to as EXPIRED	QuotationExpiryJob (Hangfire)
BR-C2-11	C2	ACCEPTED → CONVERTED only via convert-to-order endpoint — sets source_quotation_id on SO	SaleQuotationService
BR-C3-01	C3	source_type defaults to MANUAL when no quotation_id is provided	SaleOrderService
BR-C3-02	C3	source_type = FROM_QUOTATION requires source_quotation_id to be non-null	SaleOrderService
BR-C3-03	C3	When converting from quotation, source_inquiry_id is chained from quotation.source_inquiry_id	SaleOrderService
BR-C3-04	C3	customer_po_reference is free text (max 50 chars), not validated against external systems	SaleOrderService
BR-C3-05	C3	Duplicate customer_po_reference per org is allowed (warning, not error)	SaleOrderService
BR-C4-01	C4	Reserve button is DISABLED unless user has sales_reserve_inventory claim	SO Line UI
BR-C4-02	C4	Release button is DISABLED unless user has sales_release_reservation claim	SO Line UI
BR-C4-03	C4	Reserve calls IStockReservationService.Reserve with source_type=SALES_ORDER	SaleOrderReservationService
BR-C4-04	C4	Partial reservation is allowed when available_qty < requested reserve qty	SaleOrderReservationService
BR-C4-05	C4	Cancelling a Sale Order releases ALL active reservations on its lines	SaleOrderService

Rule ID	Change	Rule Description	Enforcement Point
BR-C4-06	C4	reservation_ttl_hours from SaleOrderConfig applies — expired reservations auto-released	ReservationExpiryJob
BR-C4-07	C4	Delivery indicator is COMPUTED at read time — never persisted	SaleOrderLineDto
BR-C4-08	C4	Reserve/Release buttons are visible but disabled (not hidden) for unauthorized users	SO Line UI
BR-C5-01	C5	Rejection reason codes are unique per org	RejectionReasonService
BR-C5-02	C5	Seed rejection reasons are inserted on tenant provisioning — cannot be deleted, only deactivated	TenantProvisioningService
BR-C5-03	C5	Deactivated rejection reasons are excluded from dropdowns but retained in historical data	RejectionReason query

12. Test Scenarios

Test ID	Change	Scenario	Expected Result
T-C1-01	C1	Create sale inquiry with valid customer	Inquiry created, status = RECEIVED, number auto-generated
T-C1-02	C1	Create inquiry for non-CUSTOMER partner	Validation error: partner must include CUSTOMER type
T-C1-03	C1	Add line to inquiry with product from catalog	Line created with product_id set, status = PENDING
T-C1-04	C1	Add line with free-text description (no product_id)	Line created with product_id NULL, description populated
T-C1-05	C1	Evaluate line as CAN_SUPPLY	Line status = CAN_SUPPLY, estimated_delivery_date required
T-C1-06	C1	Evaluate line as PARTIAL (qty 600 of 1000)	can_supply_quantity = 600, estimated_delivery_date required
T-C1-07	C1	Evaluate line as CANNOT_SUPPLY without rejection_reason	Validation error: rejection_reason_id required
T-C1-08	C1	Evaluate line as CANNOT_SUPPLY with rejection reason + alternative	Line saved with rejection_reason_id, alternative_product_id
T-C1-09	C1	Try to mark REVIEW_COMPLETE with PENDING lines	Validation error: all lines must be evaluated
T-C1-10	C1	Mark REVIEW_COMPLETE when all lines evaluated	Status = REVIEW_COMPLETE
T-C1-11	C1	Create quotation from reviewed inquiry	Quotation created, inquiry status = QUOTED
T-C1-12	C1	Modify line on QUOTED inquiry	Validation error: inquiry is read-only
T-C2-01	C2	Create quotation independently (no inquiry)	Quotation created, source_inquiry_id = NULL

Test ID	Change	Scenario	Expected Result
T-C2-02	C2	Create quotation from inquiry with evaluated lines	Lines pre-populated from inquiry, source_inquiry_line_id linked
T-C2-03	C2	Add REJECTED line with alternative	REJECTED line + ALTERNATIVE line with alternative_for_line_id
T-C2-04	C2	Add ALTERNATIVE without pointing to REJECTED line	Validation error: alternative_for_line_id required
T-C2-05	C2	Send quotation (DRAFT → SENT)	Status = SENT, sent_at populated
T-C2-06	C2	Try to edit line on SENT quotation	Validation error: quotation is not editable
T-C2-07	C2	Record customer ACCEPTED on a line	customer_response = ACCEPTED, date recorded
T-C2-08	C2	Record customer COUNTER without counter price	Validation error: counter_price required
T-C2-09	C2	Record customer COUNTER with price	customer_response = COUNTER, counter_price saved
T-C2-10	C2	Mark quotation accepted (at least one ACCEPTED line)	Status = ACCEPTED
T-C2-11	C2	Quotation expires past valid_to	Hangfire job sets status = EXPIRED
T-C2-12	C2	Convert accepted quotation to Sale Order	SO created with source_quotation_id, only ACCEPTED lines copied, quotation status = CONVERTED
T-C2-13	C2	Convert quotation with COUNTER lines (unresolved)	Only ACCEPTED lines in SO; COUNTER lines excluded
T-C3-01	C3	Create SO directly (no quotation)	source_type = MANUAL, source_quotation_id = NULL
T-C3-02	C3	Create SO with customer_po_reference	customer_po_reference stored, visible on header

Test ID	Change	Scenario	Expected Result
T-C3-03	C3	Attach customer PO document	Attachment created, customer_po_attachment_id linked
T-C3-04	C3	Duplicate customer_po_reference per org	Warning shown, not blocked
T-C3-05	C3	SO from quotation chains inquiry ID	source_inquiry_id = quotation.source_inquiry_id
T-C4-01	C4	View SO lines — no reservations or fulfillment	All lines show  RED indicator
T-C4-02	C4	Reserve full quantity on line (authorized user)	reserved_qty updated, indicator →  BLUE
T-C4-03	C4	Reserve partial quantity (available < ordered)	Partial reserve dialog, reserved_qty = available, indicator →  YELLOW
T-C4-04	C4	Try to reserve (unauthorized user)	Button disabled, tooltip shown
T-C4-05	C4	Release reservation (authorized user)	reserved_qty = 0, inventory returned, indicator →  RED
T-C4-06	C4	Cancel SO with active reservations	All reservations released, lines reserved_qty = 0
T-C4-07	C4	Reservation TTL expires	Hangfire releases, reserved_qty decremented
T-C4-08	C4	Line partially fulfilled (300 of 600)	indicator =  YELLOW

Test ID	Change	Scenario	Expected Result
T-C4-09	C4	Line fully fulfilled	indicator =  GREEN
T-C4-10	C4	Reserve All Open Lines (batch)	All eligible lines reserved, indicators updated
T-C5-01	C5	List rejection reasons	Returns active reasons for org, sorted by display_order
T-C5-02	C5	Create custom rejection reason	New reason created with unique code
T-C5-03	C5	Deactivate seed rejection reason	Reason deactivated, excluded from dropdowns, retained in historical data
T-C5-04	C5	Create reason with duplicate code	Validation error: code must be unique per org

13. Development Phases

13.1 Phase Breakdown

Phase	Changes	Estimated Days	Dependencies
Phase A — Schema & Lookups	C5 (RejectionReasons), M1 migrations	2 days	None
Phase B — Sale Inquiry Module	C1 (entities, services, API, UI)	5 days	Phase A
Phase C — Sale Quotation Module	C2 (entities, services, API, UI, Hangfire job)	6 days	Phase A
Phase D — Sale Order Extensions	C3 (source linking, customer PO, conversion logic)	3 days	Phase C
Phase E — Delivery Indicators & Reservation	C4 (indicators, reserve/release, RBAC, auto-release on cancel)	4 days	Phase D
Phase F — Testing & Integration	All changes — integration tests, E2E scenarios	3 days	Phase A-E
TOTAL		23 days	

13.2 Parallelization Opportunities

- Phase B (Sale Inquiry) and Phase C (Sale Quotation) can run in **parallel** after Phase A completes — they share only the RejectionReasons lookup, not each other's entities
- Phase D depends on Phase C completion (conversion logic requires quotation entities)
- Phase E depends on Phase D (reservation UI is on the Sale Order lines panel that Phase D extends)

13.3 Hangfire Jobs

Job	Schedule	Description
QuotationExpiryJob	Daily at 01:00 UTC	Scans SENT quotations past valid_to → marks EXPIRED
ReservationExpiryJob	Existing — no change	Releases reservations exceeding reservation_ttl_hours

13.4 New Permission Claims Summary

Claim Code	Module	Description
sales_inquiry_view	Sales	View sale inquiries
sales_inquiry_create	Sales	Create sale inquiries
sales_inquiry_edit	Sales	Edit inquiry lines, evaluate, change status
sales_quotation_view	Sales	View sale quotations
sales_quotation_create	Sales	Create sale quotations
sales_quotation_edit	Sales	Edit quotation lines, record customer responses
sales_quotation_send	Sales	Send quotation to customer
sales_order_create	Sales	Create sale orders (extends existing)
sales_reserve_inventory	Sales	Reserve inventory for SO lines
sales_release_reservation	Sales	Release SO line reservations

Appendix A — Entity Relationship Summary

```
lookups.RejectionReasons
  ↑ FK
demand.SaleInquiryLines
  ↑ FK
demand.SaleInquiries
  ↑ FK (source_inquiry_id)
  └──────────────────┘
                        ↓
demand.SaleOrders (extended)
                        ↓
demand.SaleOrderLines (extended: reserved_qty)
                        ↓
inventory.StockReservations (existing - SALES_ORDER source_type)
```

Appendix B — Deferred to Addendum 33

The following capabilities are explicitly NOT covered in this addendum and will be addressed in Addendum 33 — Fulfillment & Delivery:

Topic	Description
Fulfillment Orchestration	Picking lists, pack slips, fulfillment workflow
Delivery Notes	Delivery note generation, partial delivery tracking
Shipment Tracking	Carrier integration, tracking numbers, delivery proof
Invoice Generation	Invoice creation from fulfilled Sale Orders
Customer Portal	Self-service portal for order placement (source_type=PORTAL)
Inter-Tenant Orders	Cross-tenant purchase → sale order flow (source_type=INTER_TENANT)

End of FSD Addendum 32 — Sales Pre-Order Pipeline